

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	NARAPPAGARI SRAVYA	Reg. No.:	192472192
Section / Batch:	BE CSE AI / 2024	Date:	14-08-2026

Code of Conduct

*I **NARAPPAGARI SRAVYA(192472192)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: SRAVYA

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

Action: Buy

Agent: Student

Object: Book

Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.
3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudo code Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Reference Resolution Algorithm

Algorithm

Input: A discourse containing pronouns.

Output: Pronouns linked to their correct entities.

Steps

1. Read the discourse sentence by sentence.
2. Identify all entities/nouns.
3. Identify pronouns such as **he, she, it, they, him, her**.
4. Generate possible antecedents for each pronoun.
5. Check **gender and number agreement**.
6. Check the **recency** of each possible antecedent.
7. Check **semantic compatibility** with the action.
8. Select the most suitable antecedent.
9. Replace the pronoun with the selected entity.

Application

Original:

“Ravi met Arun at the library. He borrowed a book and later returned it.”

Entities:

- Ravi
- Arun
- Book

Pronouns:

- **He**
- **it**

For **He**:

- Ravi → masculine, singular
- Arun → masculine, singular

Both satisfy gender and number. However, **Ravi** is the more natural antecedent because he is the discourse subject and the next sentence describes his actions.

For **it**:

- “book” is the only suitable singular, non-human object.

Resolved Discourse

Ravi met Arun at the library. Ravi borrowed a book and later returned the book.

Validation

The resolution uses:

- **Gender:** “He” must refer to a male entity.
- **Number:** “He” refers to one person.
- **Discourse prominence:** Ravi is the main subject of the first sentence.
- **Semantic compatibility:** A person can borrow a book and return it.
- **Recency:** Recently mentioned entities are considered as possible antecedents.

Therefore, the selected references produce a coherent discourse.

2. Text Coherence and Discourse Structure Algorithm

Algorithm

Input: Multiple sentences.

Output: Discourse relations and structure.

Steps

1. Split the text into sentences.
2. Identify important entities and events.
3. Identify discourse markers such as **therefore, because, then, however**.
4. Compare consecutive sentences.
5. Identify relations such as:
 - Cause–Effect
 - Sequence
 - Elaboration

- Contrast

6. Construct a discourse structure.

7. Check whether the relations form a logical and coherent flow.

Given Discourse

“The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online.”

Discourse Relations

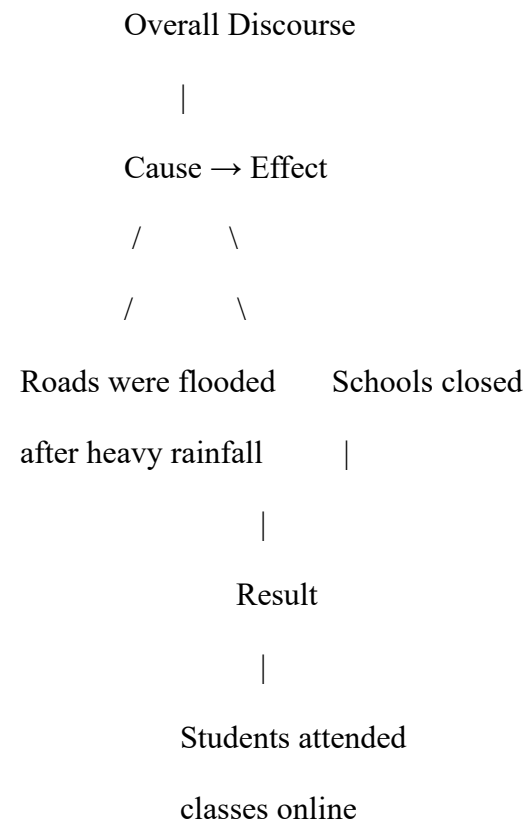
Sentence 1 → Sentence 2: Cause–Effect

- Heavy rainfall caused flooding.
- Flooded roads contributed to schools being closed.

Sentence 2 → Sentence 3: Sequence/Result

- Schools were closed.
- As a consequence, students attended classes online.

Discourse Structure



Coherence Explanation

The discourse is coherent because each sentence logically follows from the previous one:

Heavy rainfall



Roads flooded



Schools closed



Students attended classes online

The word “**Therefore**” explicitly signals the cause-effect relationship. The final sentence logically describes the consequence of school closure.

3. Dialogue Act Identification Algorithm

Algorithm

Input: Conversation between user and agent.

Output: Dialogue-act label for each utterance.

Steps

1. Read each utterance.
2. Identify its communicative purpose.
3. Detect question/request words and action statements.
4. Assign a dialogue-act label.
5. Maintain the conversation context.
6. Use the sequence of acts to determine the user's intention.

Conversation Analysis

User:

“Can you book a train ticket for me?”

Dialogue Act: Request

The user requests the agent to book a ticket.

Agent:

“Sure, where would you like to travel?”

Dialogue Act: Question

The agent asks for missing information.

User:

“I want to go to Chennai.”

Dialogue Act: Inform

The user provides the requested destination.

Agent:

“Your ticket has been booked.”

Dialogue Act: Confirmation

The agent confirms that the requested operation has been completed.

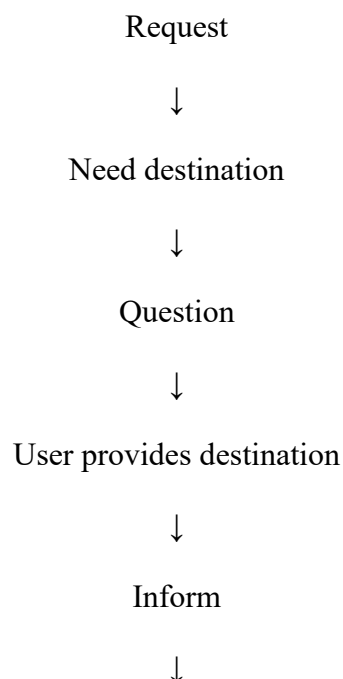
Dialogue-Act Sequence

Request → Question → Inform → Confirmation

Role of Dialogue Acts

Dialogue acts allow the conversational agent to understand the user's intention and determine what information is needed next.

For example:



Ticket booking



Confirmation

Thus, dialogue-act recognition helps maintain **context, task progression, and appropriate responses**.

4. Language Generation and Surface Realization

Semantic Input

Action = Buy

Agent = Student

Object = Book

Tense = Past

Algorithm

1. **Analyze semantic representation.**
2. Identify the agent, action, and object.
3. Perform **lexical selection**.
4. Select the appropriate tense.
5. Arrange the words according to grammatical structure.
6. Perform morphological changes.
7. Generate the final sentence.
8. Check grammatical agreement.

Step 1 – Lexical Selection

Semantic concepts are converted into words:

- Agent → **student**
- Action → **buy**
- Object → **book**

Step 2 – Tense Conversion

The input specifies **Past tense**.

Therefore:

buy → **bought**

Step 3 – Sentence Structure

For a simple English sentence:

Subject + Verb + Object

Therefore:

Student + bought + a book

Final Sentence

The student bought a book.

Surface Realization

The system adds:

- **The** → determiner
- **student** → subject
- **bought** → past-tense verb
- **a** → article
- **book** → object

Grammatical Validation

The sentence is grammatically correct because:

- It has a valid **Subject–Verb–Object** structure.
- **Bought** correctly represents the past tense of *buy*.
- The article “**a**” is correctly used before the singular count noun *book*.

5. Hybrid Interlingua + Statistical Machine Translation

Source Sentence

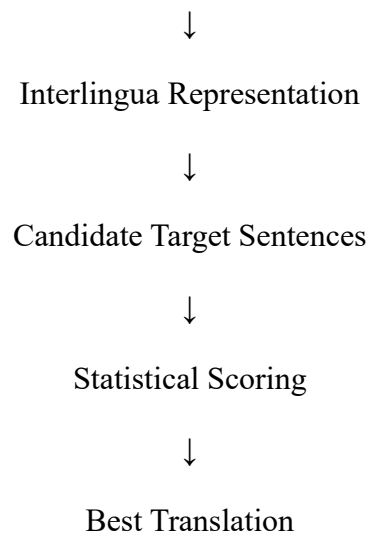
“The boy is playing football.”

Algorithm

Source Sentence



Linguistic Analysis



Step 1 – Analyze the Source Sentence

Identify:

Element Value

Subject Boy

Action Play

Object Football

Tense Present

Aspect Progressive

Number Singular

Person Third person

The sentence means that **a male child is currently playing football.**

Step 2 – Interlingua Representation

A language-independent representation can be:

PLAY(

Agent = BOY,

Object = FOOTBALL,

Tense = PRESENT,

Aspect = PROGRESSIVE

)

This representation captures the meaning independently of the target language.

Step 3 – Generate Candidate Target Translations

For example, if the target language is **Telugu**, possible candidates could include:

1. బాలుడు ఫుట్ బాల్ ఆడుతున్నాడు.
2. అబ్బాయి ఫుట్ బాల్ ఆడుతున్నాడు.

Both express approximately the same meaning, but their usage and naturalness may differ depending on context.

Step 4 – Statistical Scoring

The system evaluates candidates using factors such as:

- Translation probability
- Language-model probability
- Grammatical correctness
- Word order
- Contextual appropriateness

Conceptually:

Score(translation)

=

Translation Probability

+

Language Model Probability

+

Context Score

The candidate with the highest overall score is selected.

Step 5 – Final Translation

బాలుడు ఫుట్ బాల్ ఆడుతున్నాడు.

Meaning: The boy is playing football.

Evaluation

The **Interlingua representation** helps by separating meaning from the source language. This makes it easier to preserve important information such as:

- Agent
- Action
- Object
- Tense
- Aspect

The **statistical component** then helps choose the most natural target-language expression from multiple possible candidates.

Overall Conclusion

A hybrid approach combines the strengths of both methods:

Interlingua → preserves meaning and reduces ambiguity

Statistical translation → selects natural and probable target-language wording

Therefore, combining them can improve translation consistency and reduce errors caused by direct word-to-word translation.